John Snow LABS

Healthcare Language Models for Information Extraction

Gursev Pirge

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Healthcare Language Models

140+ million

Downloads on PyPI.
"Most Widely Used NLP Library in the Enterprise."

54% share

of GenAI in Healthcare projects at large companies use John Snow Labs

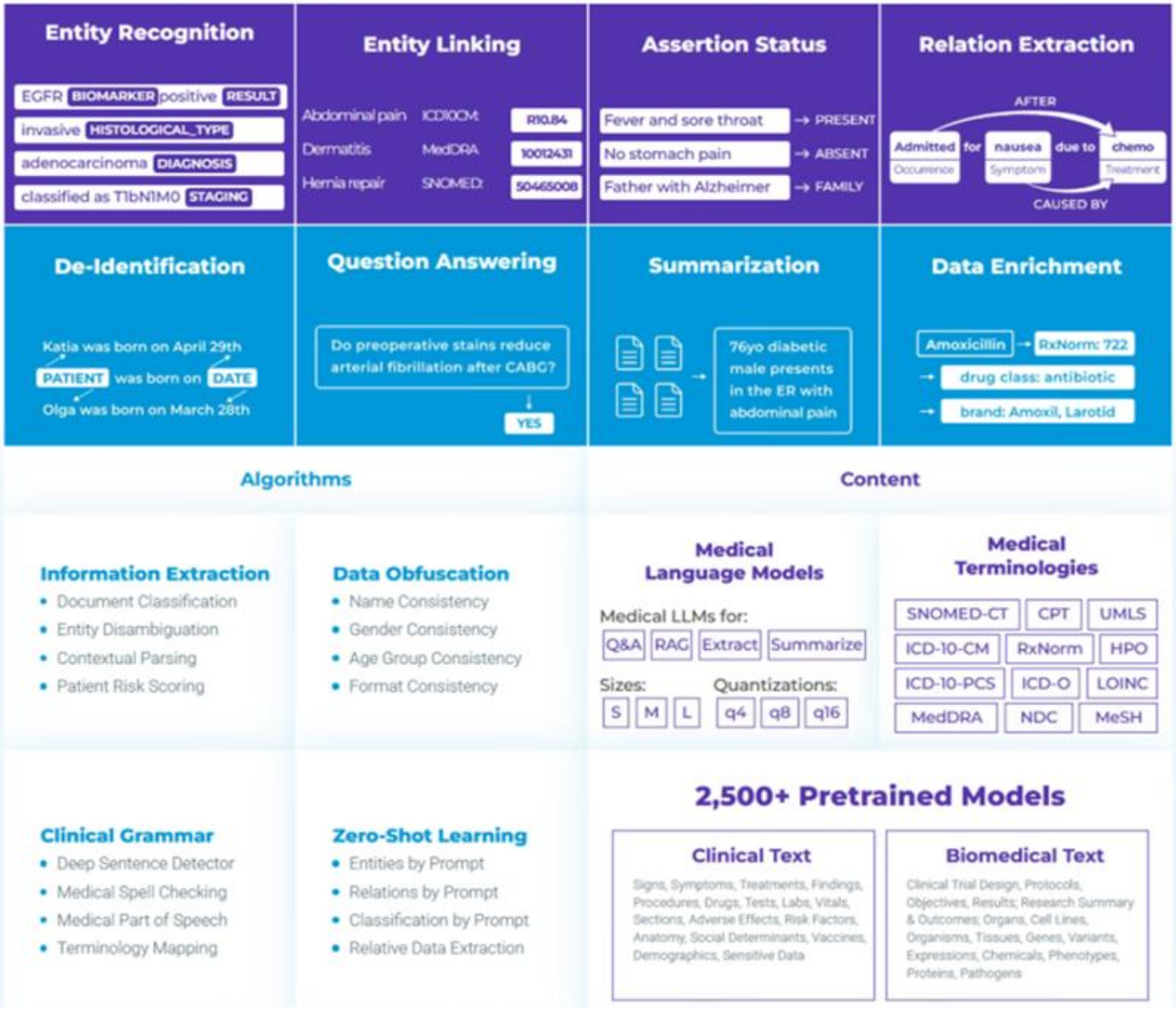
#1 Accuracy

on 25 benchmarks in peer-reviewed papers

O'Reilly Media

Gradient Flow

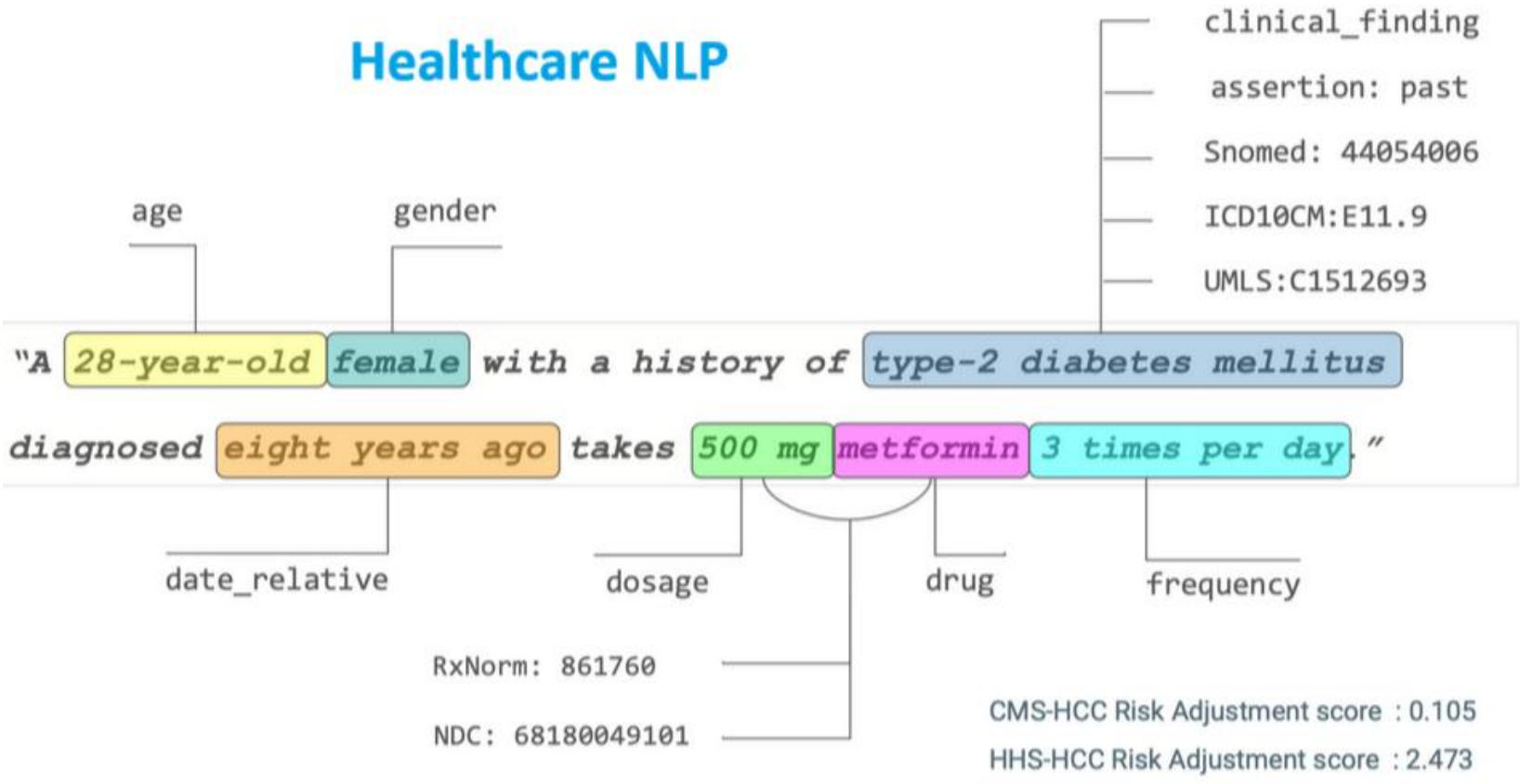
Papers with Code



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Healthcare Language Models

Named Entity Recognition



Rule-based NER

MAKE KNOWN PATTERNS AND CURATED TERMS EXPLICIT

Contextual Parser

Use surrounding context

```
context → pattern
```

Regex Matcher

Detect specified formats

```
\b\d{3}-\d{2}-\d{4}\b
```

Text Matcher

Find curated terms and phrases

```
metformin | insulin
```

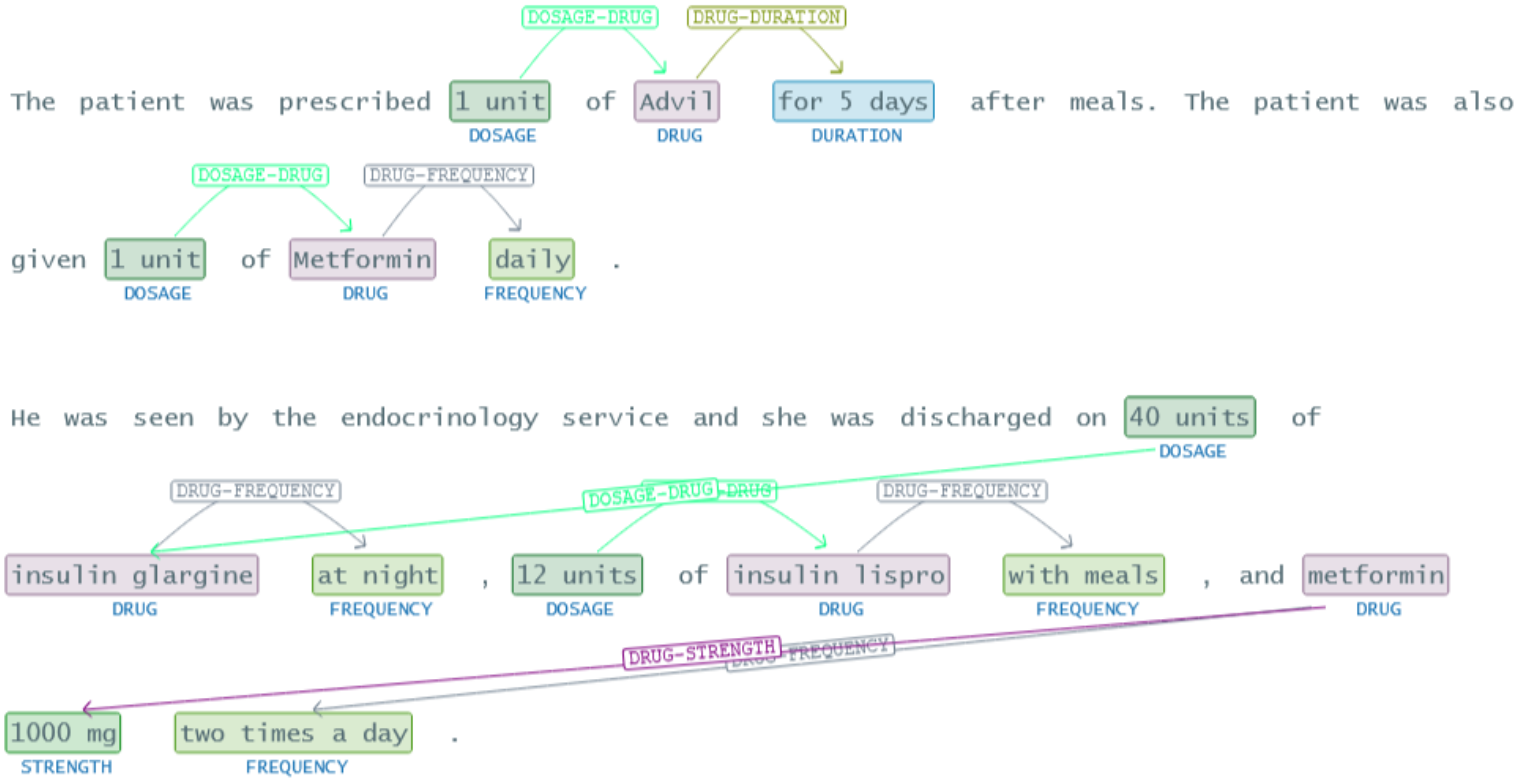
Rules complement learned recognition when coverage needs to be direct, controllable, and easy to audit.

Assertion Status Detection

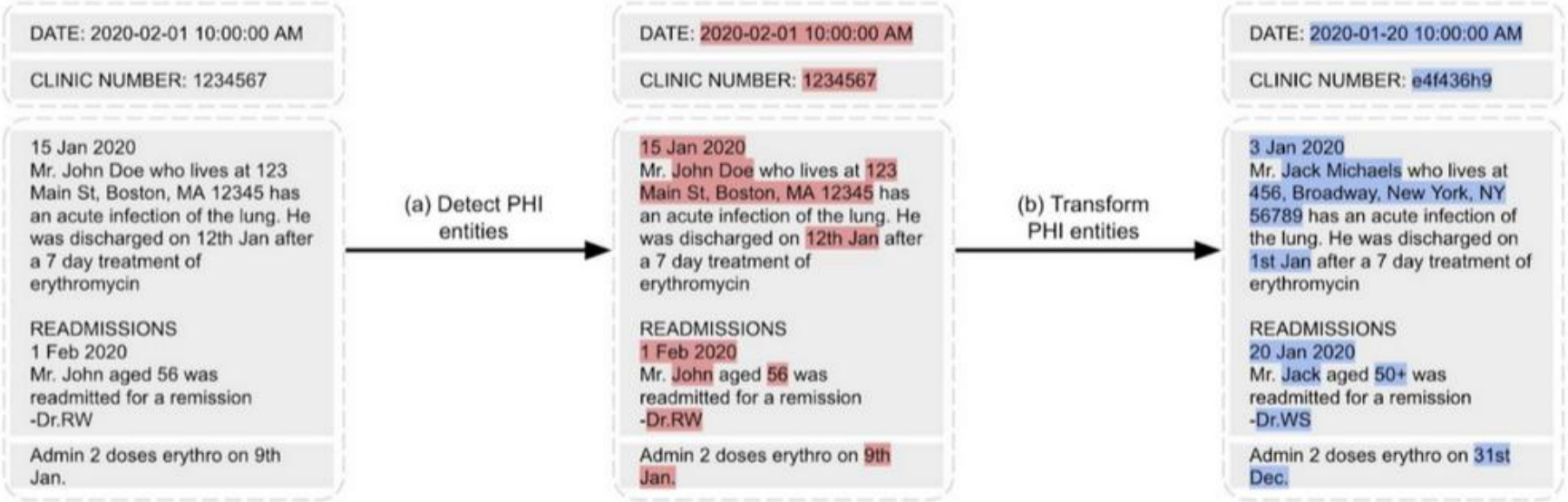
"Mother with a lung cancer, a patient is diagnosed as breast cancer in 1991 and then admitted to Mayo Clinic in Oct 2000, went under chemo for 6 months, discharged in April 2001 with a prescription of 2 mg metformin 3x per day. No sign of gynecological disorder but she suffers from acute cramps if she doesn't take her drug."

Chunk	Entity	Assertion
lung cancer	Oncological	Family
breast cancer	Oncological	Past
chemo	Treatment	Past
gynecological disorder	Disorder	Absent
acute cramps	Disorder	Conditional

Relation Extraction



Deidentification



Deidentification

APPLY THE PROTECTION MODE THAT FITS THE AUTHORIZED WORKFLOW

Privacy controls demonstrated

Masking — replace sensitive spans with protection tokens

Obfuscation — use a realistic value of the same type

Reidentification — restore original text only when permitted

Dates and structured columns — shift and protect consistently



Structured LLM outputs

DESIGN THE OUTPUT SHAPE BEFORE THE MODEL GENERATES IT

Schema-based NER

The input schema defines expected entity fields and whether each can contain one item or many.

```
Drug: [name, reaction]
Condition: text
Test: [type, result]
```

Text2SOAP

Target a familiar clinical note structure for a clear, consistent downstream result.

- S **Subjective**
- O **Objective**
- A **Assessment**
- P **Plan**

Structured output reduces ambiguity for downstream review, storage, and automation.

Multimodal PDF workflow

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BRING PDF CONTENT INTO THE SAME CLINICAL NLP WORKFLOW

1

PDF files

Documents and scans

2

Reader2Doc

Document content

3

LLM prep

Prompt context

4

MedicalLLM

Model response

5

Structured answer

Downstream use